One Result, Two Claims

Topic 6.11 · TODAY'S PROBLEM

Suppose a utility has 500,000 online accounts. It randomly selects 10,000, assigns 5,000 to message A and 5,000 to B, and records on-time payment. The pre-data question asks whether the population proportions differ at $\alpha = 0.05$; deployment requires A to improve payment by at least 5 percentage points.

Rowan: "I rejected equality, so A has proved an important improvement of at least 5 points. Deploy A."

Salma: "Rejecting equality supports a nonzero difference, but does not automatically establish the separate 5-point requirement."

On-time payment

Message	n	Paid	Other
A	5,000	3,050	1,950
B	5,000	2,950	2,050

FIRST TAKE (5 min, on your sheet)

Whose report is more defensible, Rowan's or Salma's? Cite the observed gap or the claim tested.

DOK 1 **(a)** Calculate $\hat{p}_A$, $\hat{p}_B$, and the observed difference $\hat{p}_A - \hat{p}_B$.

DOK 2 **(b)** Verify conditions and carry out the two-sided two-sample z-test of $H_0 : p_A - p_B = 0$. Compute $\hat{p}_c$, pooled SE, z, and the p-value; interpret the p-value and decide at $\alpha = 0.05$.

DOK 3 **(c)** Decide which report is warranted. Use the p-value, observed gap, 5-point standard, and null value. Explain what the rejected report would mislead a reader to believe and name the confidence level and endpoint criterion needed for deployment.

Video + follow-along: `u6_lesson11_live.html` (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.